

MD. ZAWADUL ISLAM

Electrical and Electronic Engineer

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LinkedIn: _____ | GitHub/Portfolio: _____

SUMMARY

Electrical and Electronic Engineering undergraduate with a strong foundation in power systems, electrical machines, and energy conversion, complemented by hands-on experience in robotics, embedded systems, and signal processing. Interested in applying engineering knowledge to practical challenges in the power and energy industry. Proven leadership experience organizing large-scale engineering events and managing multidisciplinary technical teams while delivering engineering solutions.

EDUCATION

Ahsanullah University of Science and Technology (AUST) **Sep 2022 - Expected** _____

BSc in Electrical & Electronic Engineering; CGPA: 3.636

Chittagong Cantonment Public College **Jun 2019 - Dec 2021**

Higher Secondary Certificate; GPA: 5.00

LEADERSHIP & ORGANIZATIONAL EXPERIENCE

Vice President, AUST Robotics Club **Jan 2026 - Present**

- Lead technical and organizational operations of the robotics club, coordinating a team of _____ members
- Direct robotics and embedded systems projects and workshops, including _____ projects/workshops
- Head content writing and social media management, planning promotional strategy and technical communications
- Supervise event publicity, announcements, and participant engagement campaigns reaching _____ participants/audience members
- Supervise technical teams for event execution, including AUST Rover Challenge 2.0, involving _____ teams/participants
- Coordinate inter-department collaboration for engineering competitions
- Manage planning, budgeting, and technical logistics of large events with budgets of approximately BDT _____

Organizing Director, AIAA AUST Student Branch **Dec 2024 - Jun 2025**

- Planned and executed technical events, seminars, and workshops in aerospace and engineering domains for _____ participants
- Managed an organizing team of _____ members, including task distribution, timelines, and event logistics
- Coordinated with technical, design, and promotion teams for smooth event delivery
- Handled participant communication, registrations, and overall event operations for _____ attendees

General Member, AUST EEE Society **Fall 2022 Semester**

PROJECTS

Autonomous Pick-and-Place Mobile Robot - Arduino

- Designed a mobile robot using Arduino _____ and _____ sensors for automatic obstacle detection and navigation
- Built a robotic arm mechanism using _____ to grip and release objects
- Programmed Arduino for fully autonomous robot operation
- Demonstrated the working principle of a simplified pick-and-place robotic system

AM Receiver Design & Implementation

- Designed and built a functional AM receiver covering approximately _____ frequency range, applying communication theory to hardware
- Tested and optimized the circuit while addressing practical issues such as noise
- Used _____ simulation/analysis tools to validate design choices and performance
- Collaborated within a team of _____ members to plan, implement, and troubleshoot the system

AR Frequency Estimation of Voice Signal - MATLAB (DSP)

- Implemented Burg, Yule-Walker, and Levinson-Durbin AR methods to model a recorded voice signal
- Estimated the first seven dominant frequencies from AR coefficients
- Compared the methods based on frequency estimation accuracy and model performance; best estimation accuracy: _____

Mole Game Simulation - MATLAB

- Built an interactive Whac-A-Mole style game using MATLAB graphics and logic
- Implemented random target appearance, scoring, and time-based control
- Applied event handling, visualization, and logical flow in MATLAB

SKILLS

Programming & Embedded: Arduino Programming, Embedded C / Microcontroller Logic |
Hardware/Platforms: _____

Engineering & Simulation: MATLAB (DSP, System Analysis), Cadence Virtuoso, Proteus, DipTrace, Tinkercad, COMSOL, AutoCAD | Electrical & Signal: Circuit Analysis & Design, Signal Processing Concepts, System Modeling

Professional: Technical Leadership, Project Management, Event Coordination, Team Management, Content Writing, MS Office & Documentation, MS Excel

ADDITIONAL INFORMATION

Languages: English, Bengali, Hindi, Urdu

Certifications: Advanced MS Word Course - Beginner to Expert (Udemy)

Awards/Activities: RoboMania 2.0 - Lead; ARC 2.0 - Segment Co-Lead; Robo Soccer, AUST CSE Carnival - 1st Runner-up

Relevant Coursework: _____

Additional Technical Details (optional): _____